

Krishna Jaikumar Kamath

872-219-6634 | kamathk@uchicago.edu | [LinkedIn](#) | [Github](#)

EDUCATION

University of Chicago

Master of Science in Computer Science

Chicago, IL

Aug 2022 - Present

Relevant Coursework: Advanced Algorithms, Parallel Programming, High Performance Computing, Big Data Application

Architecture, Distributed Systems, Cloud Computing

Birla Institute of Technology and Science

Bachelor of Engineering in Electrical and Electronics Engineering; GPA - 8.51/10

Goa, India

Aug 2016 - May 2020

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Data Mining, Optimization, Graphs and Networks, Probability and Statistics

EXPERIENCE

University of Chicago

Research Assistant

Bangalore, India

Jun 2023 - Sept 2023

- Solely developed a state-of-the-art **image extraction tool** for scanned scientific journal articles, leveraging deep learning-based optical character recognition (OCR) and image segmentation techniques.

Arcesium(D.E. Shaw & Co.)

Senior Software Engineer

Bangalore, India

Dec 2021 - Aug 2022

- Developed a **self-service platform** using **ReactJS** and **Kotlin**, empowering users to create **customized financial report templates** and **schedule report generation jobs**. This streamlined approach significantly reduced the developer effort required for template creation and revision
- Implemented various time-saving features such as **pivot tables** and **footnotes** using Kotlin and React, accelerating the template creation workflow. Successfully **onboarded multiple clients** to the platform and provided support in their initial template creation process

Software Engineer

Jul 2020 - Dec 2021

- Improved efficiency and reduced client effort during financial year-end by **enhancing multiple ETL workflows on financial data**. Leveraged Kotlin and JavaScript to optimize the workflows and achieve seamless data processing
- Developed a flexible platform in Java, JavaScript, and MyBatis to accommodate **adjustments in broker netting positions**. Ensured the reliability of the implementation through comprehensive unit and integration testing

Oyo Tech

Software Engineering Intern

Bangalore, India

Jan 2020 - Jun 2020

- Enhanced API availability and mitigated resource starvation by implementing effective **rate limiting strategies** using PHP
- Designed and implemented the user interface for a chat widget using JavaScript, seamlessly integrating it with third-party API-based services

Egnify Technologies

Software Engineering Intern

Hyderabad, India

May 2018 - Jul 2018

- Developed a **comprehensive assessment platform** using **ReactJS** and **Node.js**, empowering students to take mock entrance examinations seamlessly
- Developed ReactJS views for a file upload pipeline and storage mechanism implementation

PROJECTS

N-body simulator

- Developed an **advanced N-body simulator** in C, implementing **Barnes-Hut octree approximation**, **shared memory parallelism**, and **CUDA GPU acceleration** for significantly improved performance and scalability, achieving **real-time simulations of gravitational interactions** between a large particle systems

CPU Emulator

- Designed and implemented a **CPU emulator** with physical memory to execute algorithms like daxpy and block matrix multiplication.
- Engineered a **dynamic cache system** with adjustable RAM, cache, and block sizes, supporting various placement strategies (direct mapped, fully associative, n-way), block replacement policies, and write policies.
- Conducted extensive performance analysis, optimizing cache configurations to minimize instruction counts, read/write hits, and misses, resulting in enhanced computational efficiency.

Optimization Approaches for the Traveling Salesman Problem

- Implemented **Ant Colony Optimization** and **2-opt** algorithm in C++ to find approximate solutions to the classic travelling salesman problem

TECHNICAL SKILLS

Languages: Java, C/C++, Python, ReactJS, JavaScript, Go

Tools: Git, SQL, OpenMP, MPI, LaTeX